**Question 1** | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| s1 | | a | g | **g** | **t** | **a** | **b** |
| s2 | **g** | x | **t** | x | **a** | y | **b** |

**The length is 4**

Solveing it using Dynamic Programming

**For example:**

| Input | Result |
|---|---|
| aab<br>azb | 2 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>

int max(int a, int b)
{
    return (a > b) ? a : b;
}

int main()
{
    char s1[100], s2[100];
    int i, j;

    scanf("%s", s1);
    scanf("%s", s2);

    int m = strlen(s1);
    int n = strlen(s2);

    int dp[m + 1][n + 1];

    for (i = 0; i <= m; i++)
    {
        for (j = 0; j <= n; j++)
        {
            if (i == 0 || j == 0)
                dp[i][j] = 0;

            else if (s1[i - 1] == s2[j - 1])
                dp[i][j] = dp[i - 1][j - 1] + 1;

            else
                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1]);
        }
    }

    printf("%d", dp[m][n]);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | aab<br>azb | 2 | 2 | ✔ |
| ✔ | ABCD<br>ABCD | 4 | 4 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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